

Lecture 2

Solution-Solubility

الرؤية لكلية الصيدلة الجامعة الحديثة:

"تتطلع كلية الصيدلة-الجامعة الحديثة للتكنولوجيا والمعلومات إلى التميز والابتكار في مجالات التعليم الصيدلي والبحث العلمي وخدمة المجتمع وتنمية البيئة محليا وإقليميا تحت مظلة أهداف ورؤية مصر ٢٠٣٠ للتنمية المستدامة"

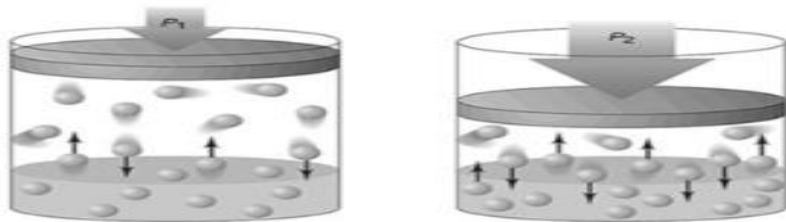
الرسالة لكلية الصيدلة الجامعة الحديثة :

"تلتزم كلية الصيدلة بالجامعة الحديثة للتكنولوجيا والمعلومات بتقديم برامج أكاديمية متطورة في التعليم الصيدلي لأعداد خريج متميز قادر علي تطوير منظومة الرعاية الصحية والارتقاء بالخدمات الصيدلية والتصنيع الدوائي وتحقيق مشاركة فعالة في مجال البحث العلمي وخدمة المجتمع لتحقيق أهداف التنمية المستدامة في إطار من القيم والأخلاقيات."

Solutions can exist in all three physical states

Example	solvent	Solute
Metal alloys السبائك المعدنية مثل النحاس الأصفر (خليط من النحاس والزنك)	Solid	Solid
Sugar solution	Liquid	Solid
Hydrogen in platinum	Solid	Gas
Alcoholic beverages	Liquid	Liquid
Carbonated water	Liquid	Gas
Air	Gas	Gas

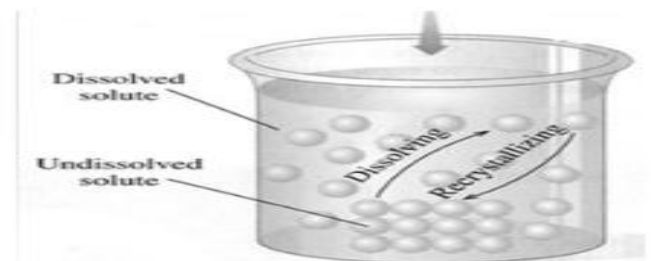
gas/liquid



liquid/liquid



solid /liquid



Solubility

Definition: the maximum amount of the solute that can dissolve in a given amount of solvent

- Importance of studying the phenomenon of solubility
 1. Select the best solvent for a drug or a mixture of drugs.
 2. Overcome problems arising during the preparation of pharmaceutical solutions.
 3. Have information about the structure and intermolecular forces of the drug.
 4. Assess purity of the drug.
 5. Qualitatively and quantitatively analyze the drug in the dosage form.

Solubility

Solubility Expression:

- Qualitative Expression **تعبيرات كيفية**
- Quantitative Expression **تعبيرات كمية**

Solubility Expression

✓ Number of milliliters of solvent in which **1 gram** of solute dissolves

1. Qualitative Expression

Descriptive term	Parts of solvent required for 1 part of solute (1 :X)
Very soluble	Less than 1
Freely soluble	From 1 - 10
soluble	From 10 - 30
Sparingly soluble	From 30 - 100
Slightly soluble	From 100 - 1000
Very slightly soluble	From 1000 - 10,000
Practically insoluble	More than or equal to 10,000

2.Quantitative Expression

A. Concentration by Percent

B. Concentration in terms of Moles

A. Concentration by Percent:

➤ **Percent concentration by mass**

(% w/w)

➤ **Percent concentration by volume**

(%v/v)

(%w/v)

➤ **A 5% w/w of NaCl solution**

means 5 g sodium chloride 100 g solution

A. Concentration by Percent:

- If both solute and solvent are liquids, the concentration may be expressed as **percent by volume (v/v)**.
- Both ethyl alcohol and water are liquids; the concentration of alcohol-water solutions is often given as percent by volume.
- **For example, a 95% v/v solution of ethyl alcohol**
Means 95 mL ethyl alcohol 100 mL solution
- **For example, a 10% w/v solution of sodium chloride**
Means 10 g sodium chloride 100 mL solution

B. Concentration in terms of Moles:

- **Molarity**
- **Molality**
- **Mole Fraction**

B. Concentration in terms of Moles:

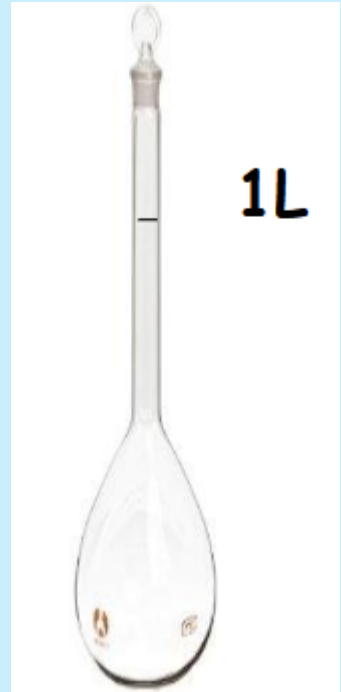
❖ **Molarity**: It is the number of moles of solute per liter of solution.

$$\text{Molarity} = \frac{\text{Number of moles}}{\text{Volume of solution in liter}}$$

$$1\text{M NaOH} = \frac{1 \text{ mole NaOH}}{\text{Volume of solution in liter}}$$

$$\text{No of moles} = \frac{\text{weight}}{\text{Molecular weight}}$$

If 1 mole, therefore weight = Molecular weight



B. Concentration in terms of Moles:

❖ Molality: It is the number of moles of solute per Kg of solvent.

$$\text{Molality} = \frac{\text{Number of moles}}{\text{Weight of solvent in Kg}}$$



B. Concentration in terms of Moles:

❖ Mole Fraction: The mole fraction of a solution component is defined as the moles of the components divided by the total moles of all components present in the solution.

❖ If we have two components A and B

❖ Mole Fraction of A =
$$\frac{\text{Number of moles of A}}{\text{Sum of no.of moles of all components}}$$

(No. of moles of A + No. of moles of B)

❖ Mole % of A = Mole Fraction of A X 100

Solubility expression summary

% w/w	No of grams of solute per 100 grams of solution	$\frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$
% w/v	No of grams of solute per 100 milliliters of solution	$\frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$
% v/v	No of milliliters of solute per 100 milliliters of solution	$\frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$

Solubility expression summary

Molarity	No of moles of solute per 1 liter of solution	$\frac{\text{Number of moles}}{\text{Volume of solution in liter}}$
Molality	No of moles of solute per 1000 grams of solvent	$\frac{\text{Number of moles}}{\text{Weight of solvent in Kg}}$
Mole Fraction	$\frac{\text{No. of moles A}}{\text{Sum of no. of moles of all components}}$	Mol%= mol fraction X 100

Methods For Determination of Solubility



**Analytical
Method**

**Synthetic
Method**

**Modified
Synthetic
Method**

**Phase
Solubility
Analysis**

The following points should be considered before determination of solubility:

[1] The solvent and solute must be pure because impurities affect solubility.

[2] Temperature should be accurately controlled.

[3] A saturated solution must be obtained before any solution is removed for analysis.

[4] Method of separation of saturated solution from the undissolved solid must be satisfactory.

[5] Accurate and reliable method of analysis of the saturated solution should be used.

Method for determination of solubility :

1. Analytical Method : It is the official method.

Uses techniques like **spectroscopy, chromatography, or gravimetric analysis** to determine solubility.

2. Synthetic Method :

Involves dissolving a known **amount of solute in a solvent and measuring the amount that remains undissolved.**

Method for determination of solubility :

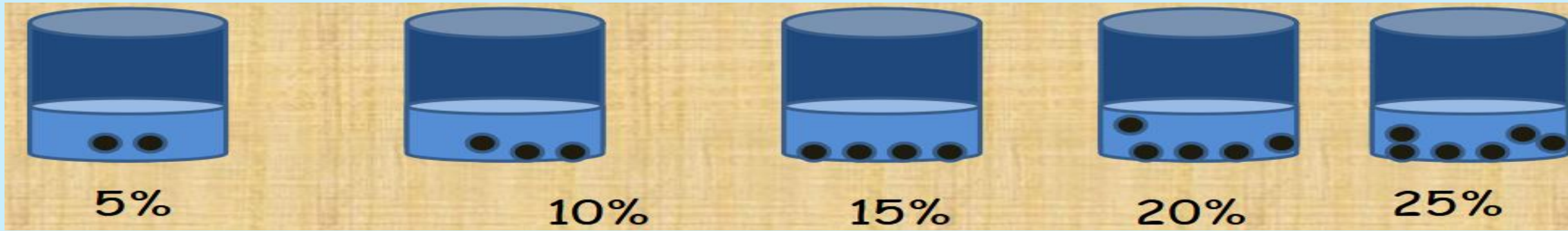
3. Modified Synthetic Method:

A variation of the synthetic method that incorporates modifications to improve accuracy or adapt to specific solutes.

4. Phase Solubility Analysis:

Examines the solubility of a compound under different conditions, such as **pH or temperature, to determine phase equilibrium.**

Phase Solubility Analysis: 6 steps

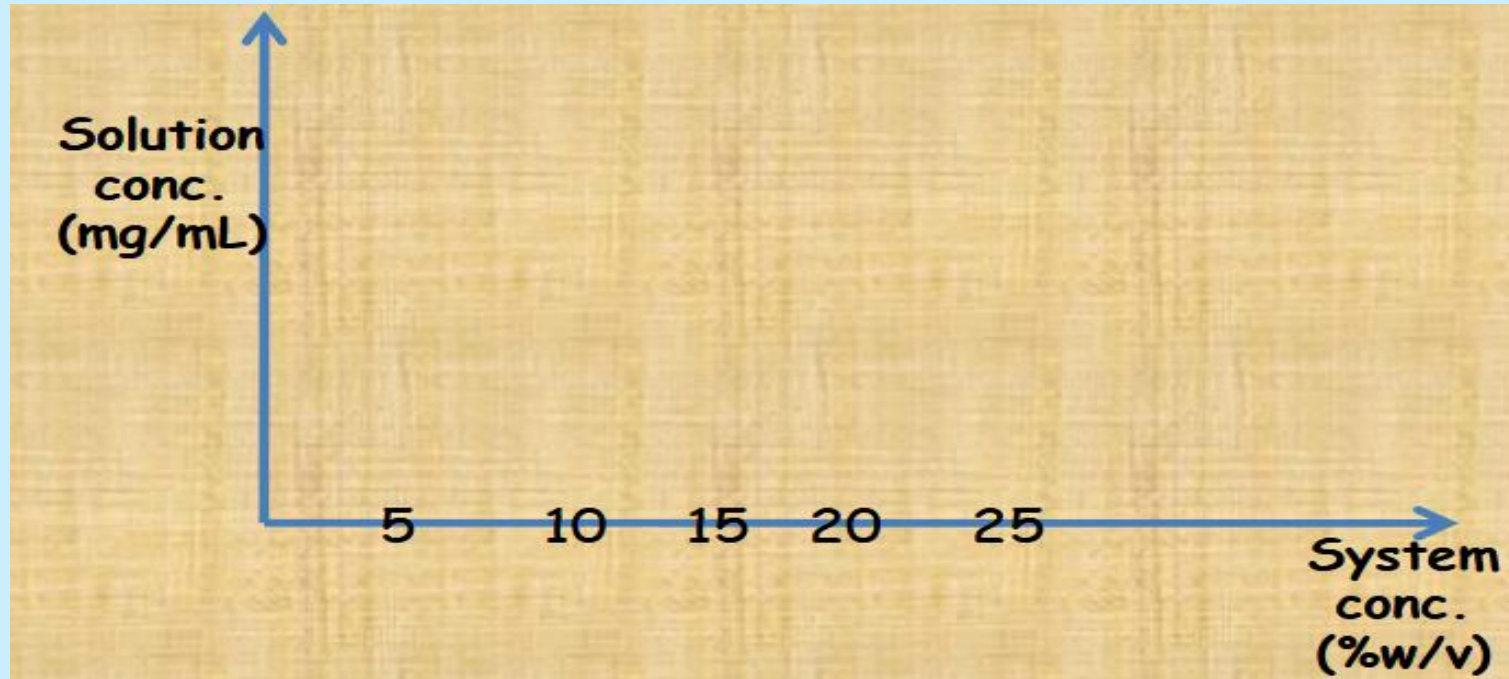


1. Mixing increasing amounts of substance with known amounts of solvent.
2. Establishment of equilibrium for each system at constant temperature.
3. Separation of the solid phase from solutions.
4. Determination of the concentration of the material dissolved in the various solutions.

Phase Solubility Analysis: 6 steps

5. Plotting amount of dissolved material per unit volume of solvent (solution concentration) against mass of material per unit volume of solvent (System Concentration).

6. Extrapolation and Calculation

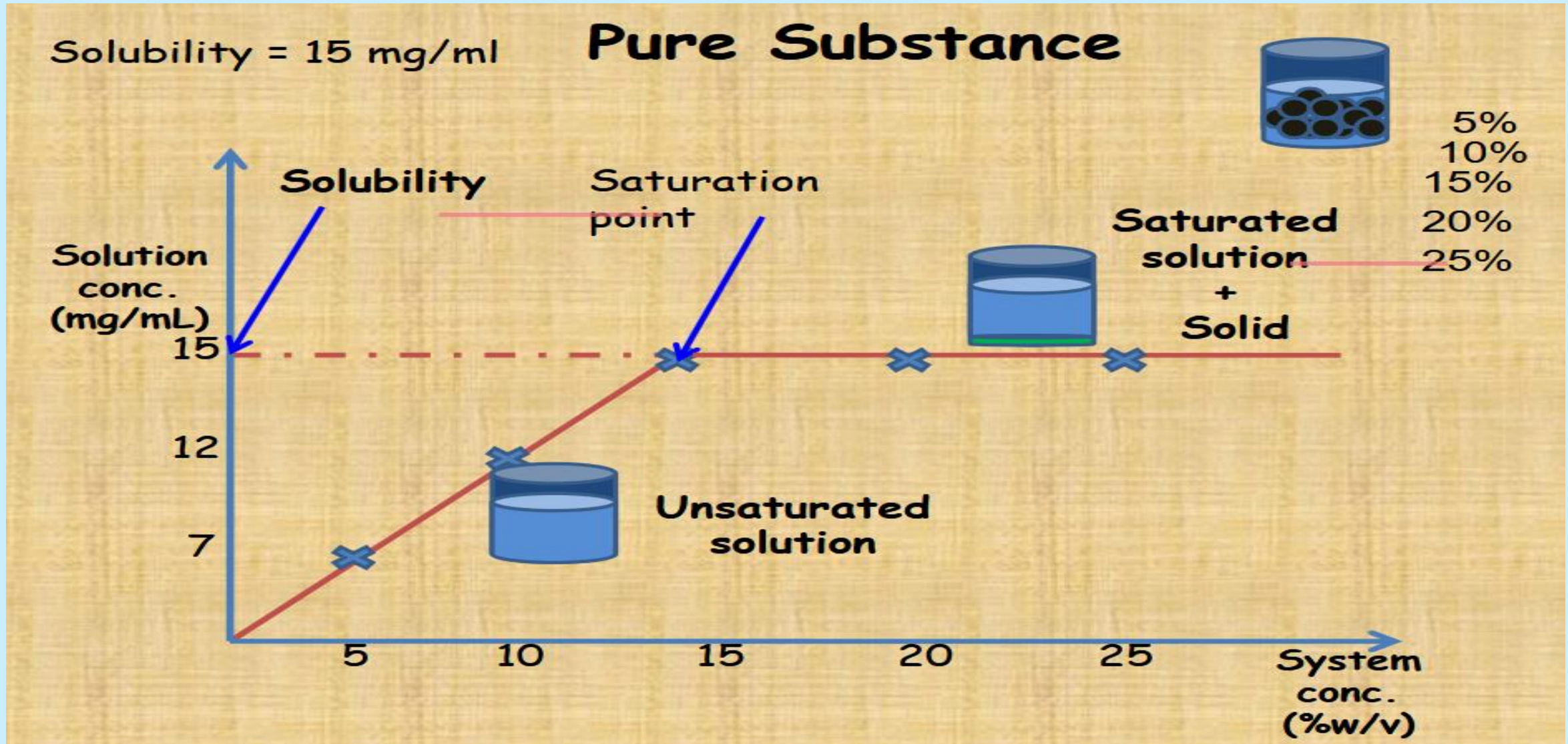


Phase Solubility Analysis:

- **For Pure Substance**
- **For impure Substance**

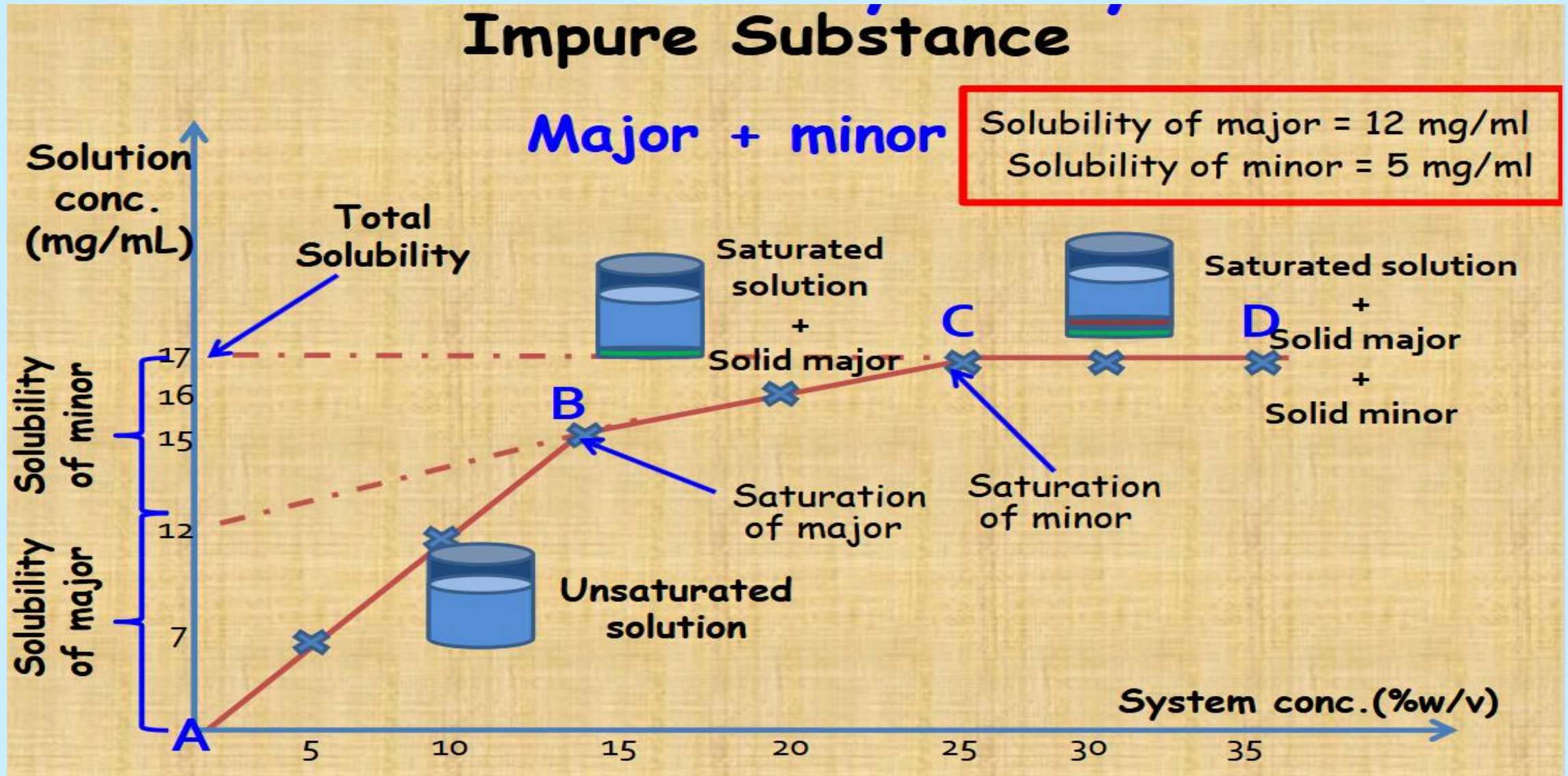
Phase Solubility Analysis:

➤ For Pure Substance



Phase Solubility Analysis:

➤ For impure Substance



Phase Solubility Analysis:

Advantages:

1. Most useful and accurate method for determination of purity of substance.
2. Can be used to obtain exact solubility of pure substance without necessity to be pure.
3. Can be used for purification of drugs $\Rightarrow$ by filtration at BC
4. Can be used to obtain fraction of impurity $\Rightarrow$ slope of BC (% purity)
5. Applicable to all species of molecules
6. Sensitive $\Rightarrow$ distinguish between optical isomers unless their ratio is 1:1

Difference between solubility and Dissolution

Solubility	Dissolution
It is the concentration of substance in saturated solution at definite temperature.	It is the rate of transfer of molecules or ions from a solid state into solution.
It is a dynamic process	It is a kinetic process.

References :

- Florence, Alexander T; Attwood, David 2015, Physicochemical principles of pharmacy Australian prescriber.
- Blaine Templar Smith 2016 Remington: The Science and Practice of Pharmacy Pharmaceutical Press.
- Ram I. Mahato, Ajit S. Narang, 2017 Pharmaceutical Dosage Forms and Drug Delivery 3rd ed, Pharmaceutical Press

Thank you

The image features the words "Thank you" in a 3D isometric font. The letters are light yellow with blue outlines and are arranged in two rows: "Thank" on top and "you" below it. Each letter casts a soft red shadow onto the pink background. Scattered around the text are several colorful, five-pointed stars in shades of cyan, yellow, and purple, each also casting a red shadow. The overall style is bright, cheerful, and modern.